

Project Presentation Master Deck

LearnLens AI: "Show your AI what you are learning."

Slide 1: Title & Tagline

- **Title:** LearnLens AI
 - **Tagline:** "Show your AI what you are learning."
 - **Subtitle:** A Local-First Multimodal AI Learning Environment & Personal Teacher
 - **Speaker Notes:** "Welcome. Today we introduce LearnLens AI. The fundamental philosophy is simple: Don't make the human adapt to the AI. Make the AI understand what the human is learning."
-

Slide 2: The Core Problem: Modern Learning is Broken

- **Bullet Points:**
 - Millions of hours wasted on passive video watching.
 - Core technical knowledge is hidden across fleeting audio and complex visual diagrams.
 - Manual note-taking disrupts learning flow and retention.
 - Existing video summarizers process only text transcripts, ignoring architecture diagrams, equations, and code.
 - **Speaker Notes:** "When students and professionals study on YouTube, Coursera, or Fortinet, they spend hours pausing, rewinding, and manually taking notes. Critical diagrams are easily forgotten."
-

Slide 3: Our Solution: One AI That Learns Alongside You

- **Bullet Points:**
 - Grant explicit observation access to any active Chrome tab.
 - The AI observes speech, video frames, slide decks, diagrams, and OCR text in real time.
 - Automatically compiles a private, session-scoped knowledge base.
 - Becomes an interactive personal tutor grounded strictly in what was taught.
 - **Speaker Notes:** "Instead of copying URLs into a generic summarizer, you simply show your AI what you are studying. It watches, understands, and remembers."
-

Slide 4: System Architecture & Workflow

- **Workflow Pipeline:**
`Chrome Tab -> Audio/Visual Capture -> faster-whisper + OCR + dHash -> Multimodal Fusion -> Session RAG -> AI Teacher -> PPT / PDF / Practice Quiz`
 - **Speaker Notes:** "Our architecture cleanly separates perception, knowledge storage, and pedagogical delivery. Everything runs locally on the user's computer."
-

Slide 5: Platform-Independent Universal Capture

- **Bullet Points:**
 - Operates on any user-authorized Chrome source: Coursera, Fortinet Training Institute, YouTube, Forage, Udemy, online lectures, or local video files.

- Utilizes native browser permissions (like Google Meet).
 - No fragile platform-specific web scrapers.
 - **Speaker Notes:** "By using the browser's native screen sharing abstraction, LearnLens AI works universally across every educational website without requiring special integrations."
-

Slide 6: Multimodal Understanding Beyond Text

- **Bullet Points:**
 - Audio: Timestamped speech transcription with speaker identification.
 - Screen: Visual structure, slide changes, and layout understanding.
 - Vision: Categorization into DIAGRAM, SLIDE, CODE, TABLE, or DEFINITION.
 - OCR: Instant extraction of slide bullets, technical rules, and labels.
 - **Speaker Notes:** "When an instructor presents a firewall diagram, LearnLens AI doesn't just hear the words—it analyzes the visual topology and connects it directly to the lecture audio."
-

Slide 7: Intelligent Frame Deduplication Engine

- **Bullet Points:**
 - Perceptual difference hashing (dHash) detects meaningful scene changes.
 - Discards 85%+ redundant or stationary video frames.
 - Extreme efficiency: zero CPU bottleneck on standard student laptops.
 - **Speaker Notes:** "We don't blindly capture a screenshot every 5 seconds. Our perceptual hashing engine ensures that only unique, high-value learning evidence is preserved."
-

Slide 8: Interactive Personal AI Teacher

- **Bullet Points:**
 - 12 Pedagogical Modes: Simple, Detailed, Exam Focus, Quick Revision, Example, Teach From Scratch, Active Recall, Flashcards, Practice Quiz, Weak Areas, Compare Concepts, Ask Anything.
 - Grounded in source timestamps with clickable evidence thumbnails.
 - Strict anti-hallucination protocol.
 - **Speaker Notes:** "The AI Teacher answers like a world-class tutor. Ask for an analogy, an exam cheat-sheet, or a deep dive, and it cites the exact minute in the lecture where the instructor explained it."
-

Slide 9: Session-Scoped Knowledge Isolation

- **Bullet Points:**
 - Dedicated knowledge namespace per learning session.
 - Fortinet Network Security never contaminates an unrelated Machine Learning course.
 - Instant vector retrieval under 5 milliseconds.
 - **Speaker Notes:** "One of our most important innovations is session isolation. The AI maintains clean, segregated memory for each course you take."
-

Slide 10: Automated PowerPoint (PPT) Generation

- **Bullet Points:**

- One-click native 16:9 widescreen presentation generation via ``python-pptx``.
 - 5 customizable styles: Teaching, Exam Revision, Quick Summary, Detailed, Visual.
 - Embeds actual captured diagrams and structured definition cards.
 - **Speaker Notes:** "Our PPT generator never dumps raw transcripts onto slides. It plans an agenda, structures definitions, explains mechanisms, and embeds the actual captured lecture diagrams."
-

Slide 11: Publication-Grade PDF Study Packs

- **Bullet Points:**
 - PDF A: Visual Study Pack (chronological slide images, OCR text, timestamps).
 - PDF B: AI Teaching Report (executive summary, definitions table, mechanisms, exam tips, and self-check questions).
 - **Speaker Notes:** "Two tailored study documents: a visual deck for quick visual review, and an in-depth teaching report formatted with ReportLab."
-

Slide 12: Ethical Assessment Boundary

- **Bullet Points:**
 - Ingests quiz PDFs and extracts practice questions.
 - Diagnoses knowledge gaps and explains correct/incorrect choices.
 - **Zero automated test-taking:** The human learner remains in full control.
 - **Speaker Notes:** "Our philosophy is unequivocal: AI prepares the human; the human controls the final action. We help students master the concepts without violating academic integrity."
-

Slide 13: First-Class Human Control

- **Bullet Points:**
 - Prominent ``[ PAUSE AGENT ]``, ``[ RESUME ]``, ``[ STOP ]``, and ``[ COMMAND ]`` controls.
 - Natural language commands: 'Generate PPT', 'Make notes', 'Explain what I learned'.
 - Human can interrupt the agent at any point.
 - **Speaker Notes:** "The human learner is always in the driver's seat. You can pause the agent, ask a clarifying question, issue new commands, and resume without missing a beat."
-

Slide 14: Local-First Privacy Sovereignty

- **Bullet Points:**
 - Runs locally: All frames, transcripts, and database records reside on your machine.
 - Zero access to browser passwords, cookies, or personal authentication tokens.
 - Total one-click data erasure.
 - **Speaker Notes:** "Privacy is paramount. Your learning history and screen content never leave your laptop."
-

Slide 15: Conclusion & Summary

- **Title:** LearnLens AI
- **Tagline:** *"Show your AI what you are learning."*
- **Key Takeaways:**

- Universal Chrome observation.
- Multimodal understanding.
- Session-scoped grounded tutor.
- Instant PPT and PDF generation.
- Complete human control.
- **Speaker Notes:** "LearnLens AI transforms how we learn from digital media. Thank you."